

02LQD Specification and Simulation of Digital Systems
2021/02/19

1. Combinational circuit design (2 points)

A museum has three rooms, each with a motion sensor (m0, m1, and m2) that outputs 1 when motion is detected. At night, the only person in the museum is one security guard who walks from room to room. Create a circuit that detects whether the guard is properly patrolling the museum, detected by *exactly* one motion sensor being 1 (by setting an output A to 1). If no motion sensor is 1, the guard may be sitting, sleeping, or absent.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z. The output is the same as the input was three clock periods previously. The first three values of Z are 0.

Example:

X = 0 1 0 1 1 0 1 0 1 1 0 1 0 0 0 1

Z = 0 0 0 0 1 0 1 1 0 1 0 1 1 0 1 0

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Draw the state diagram of the Moore Machine (**4 points**)
3. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
4. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 16-bit input Inp, a 1-bit input Go, a 1-bit output Finish, an x-bit output Counter (x to be defined by the candidate) and the signals (to be defined by the candidate) required to access the RAM. The functional unit should read the RAM and output the number of cells whose value equals Inp. Operation starts when Go is asserted and Finish is set to 1 when operation is over and the result is available on Counter. Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Monday February 22 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.